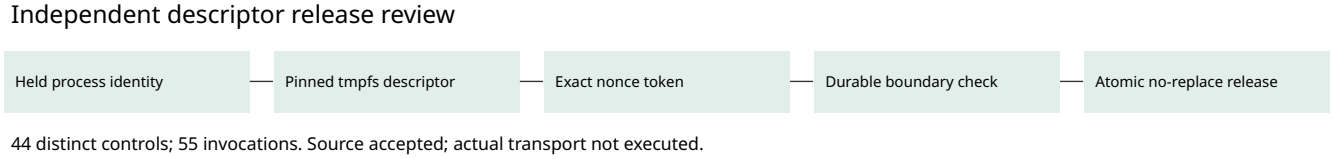
Independent descriptor release source review

Accepted the corrected descriptor release implementation at its source/software scope. One bounded independent wrapper passed 55 invocations representing 44 distinct controls; 11 invocations repeat inherited tests. Three reviewer controls cover bounded descriptor reads and symlink rejection, unsupported ABI rejection before libc access, and the single kernel-root exception to ordinary NOFOLLOW opens.



The reviewer authored the enclosing private supervisor, but not this release writer. This is a nonblind integration review. The review plan was frozen before reading the candidate implementation, using the proposed design and primary Linux documentation. The exact corrected source and tests are bound in execution-freeze.json and review.json; all input hashes remained unchanged during replay.

Two findings were corrected before independent replay. First, repeatedly truncating transport.json could destroy the durable prepublication observation after rename. The implementation now writes exclusive transport-001/002/003 records, fsyncs each file and the receipt directory, and retains the prepublication record. Second, receipt I/O followed the last boundary check. The corrected sequence repeats signal, deadline, endpoint, process and mount-view validation after receipt persistence immediately before the sole rename attempt.

The source pins a decimal host PID through a held proc directory and pidfd, verifies start ticks and cgroup identity, and deliberately follows only the kernel proc-root link. Work is opened relative to that held root with NOFOLLOW. Source checks require readonly root, root-owned mode 1777 work directory, writable tmpfs with nosuid/nodev/noexec and at most 16 MiB capacity. Reopened and held work descriptors must match device and inode.

The token is an exclusively created regular file with 37 canonical UUID/newline bytes, mode 0444 after explicit fchmod, one link and root ownership. Descriptor content and EOF are checked. Linux renameat2 with RENAME_NOREPLACE publishes once. All final-name collisions fail without overwriting. Post-attempt errors retain possible-release uncertainty; the final name is never deleted as rollback. Staging cleanup is restricted to the matching held inode. The fast-exit case remains subject to subsequent case-result checks rather than a post-rename liveness requirement.

The single independent wrapper used 2 GiB address space and a 120-second outer ceiling. Mock proc, pidfd and mount observations were combined with real unprivileged temporary-file operations; UID/GID custody was mocked where required. No proc write, container, daemon, mount, service, scientific worker or new runtime was launched. The actual process-root permission and tmpfs transport remain a separately admitted qualification dependency. The source assumes the pinned gate and fixed mount controls; it does not freeze process lifetime or defend against an unrelated privileged host actor.

Primary documentation: https://man7.org/linux/man-pages/man2/pidfd_open.2.html ; https://man7.org/linux/man-pages/man5/proc_pid_root.5.html ; https://man7.org/linux/man-pages/man2/renameat2.2.html . Local glibc statfs and statvfs headers were checked for the Linux x86-64 ABI and flags. This review does not authorize runtime execution or change any budget.